The Swarm Robotics Simulator

Simulation of the world and its dynamics

This part **knows everything** and **has access to everything** it needs.

Simulation from a robot's point of view (pretending that we really have an autonomous physical robot out in the world)

This part **doesn't know anything** (not even about the robot itself) outside of the robot's control program, unless there is a **sensor** or message communicated to the robot that provides this information.

It also **has no access to anything** (even to the parts of the robot itself), unless there is a control command available for that.

Your controller goes here!

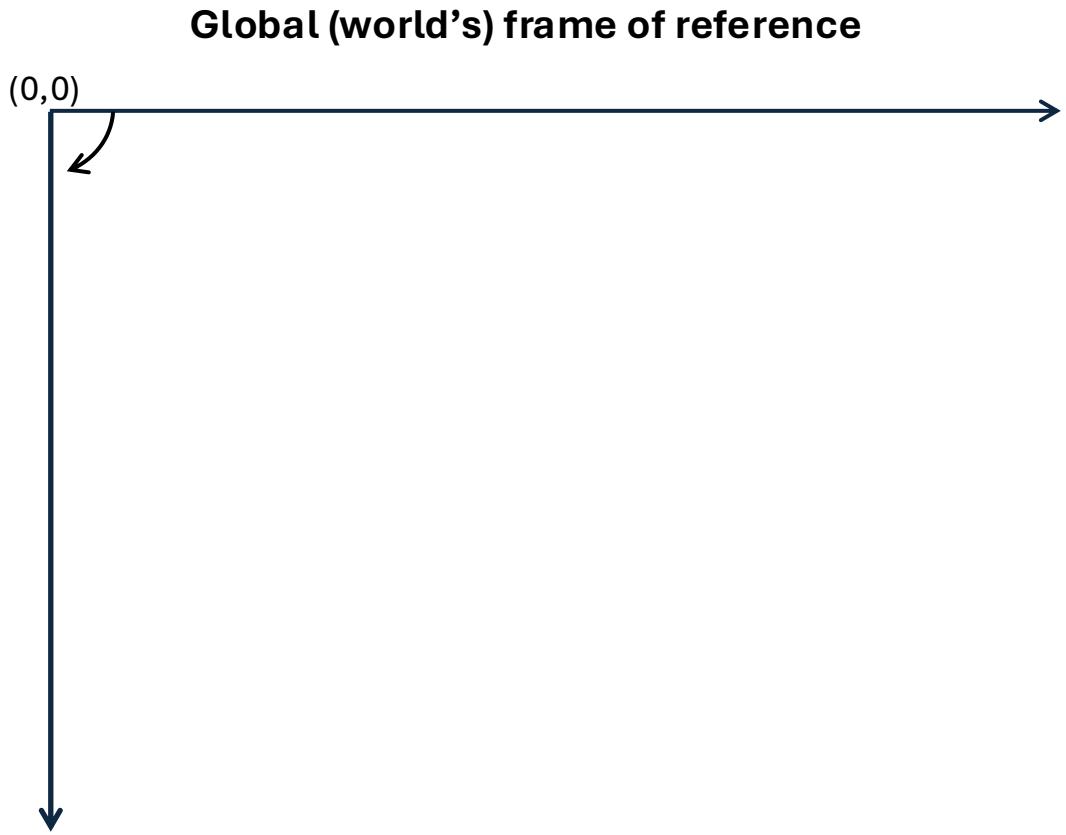
The Swarm Robotics Simulator

Simulation of the world and its dynamics

World's frame of reference
World's coordinates (the 800x600 pixel world)

All the positions and orientations (**in world's frame**):
Robots' positions (x,y) & headings
Walls, obstacles, light sources, etc.

Dynamics of everything, e.g.:
how things move?
if and how things collide?
what value the robots' sensors return considering the environment, its content, and the robots' position and orientation?



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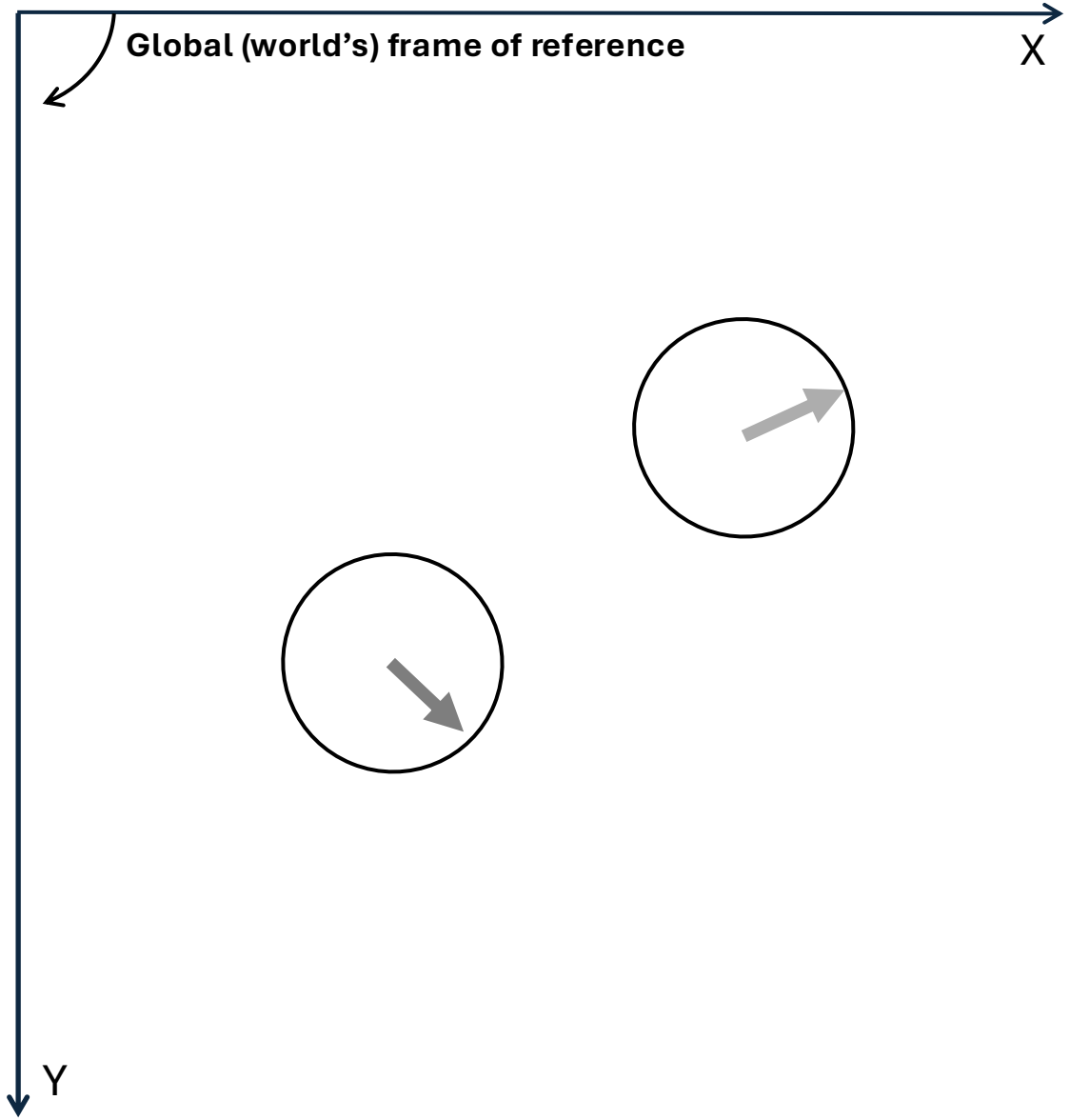
Sensor values & received messages:
- IMU: robot 's own heading → **World's frame** of reference
- Proximity/RAB: **Distance** to the detected object/robot,
- Proximity/RAB: **Bearing** of the detected object/robot – angle relative to robot's heading. → **Robot's frame** of reference (for a robot, zero angle is along its heading)
- …

…

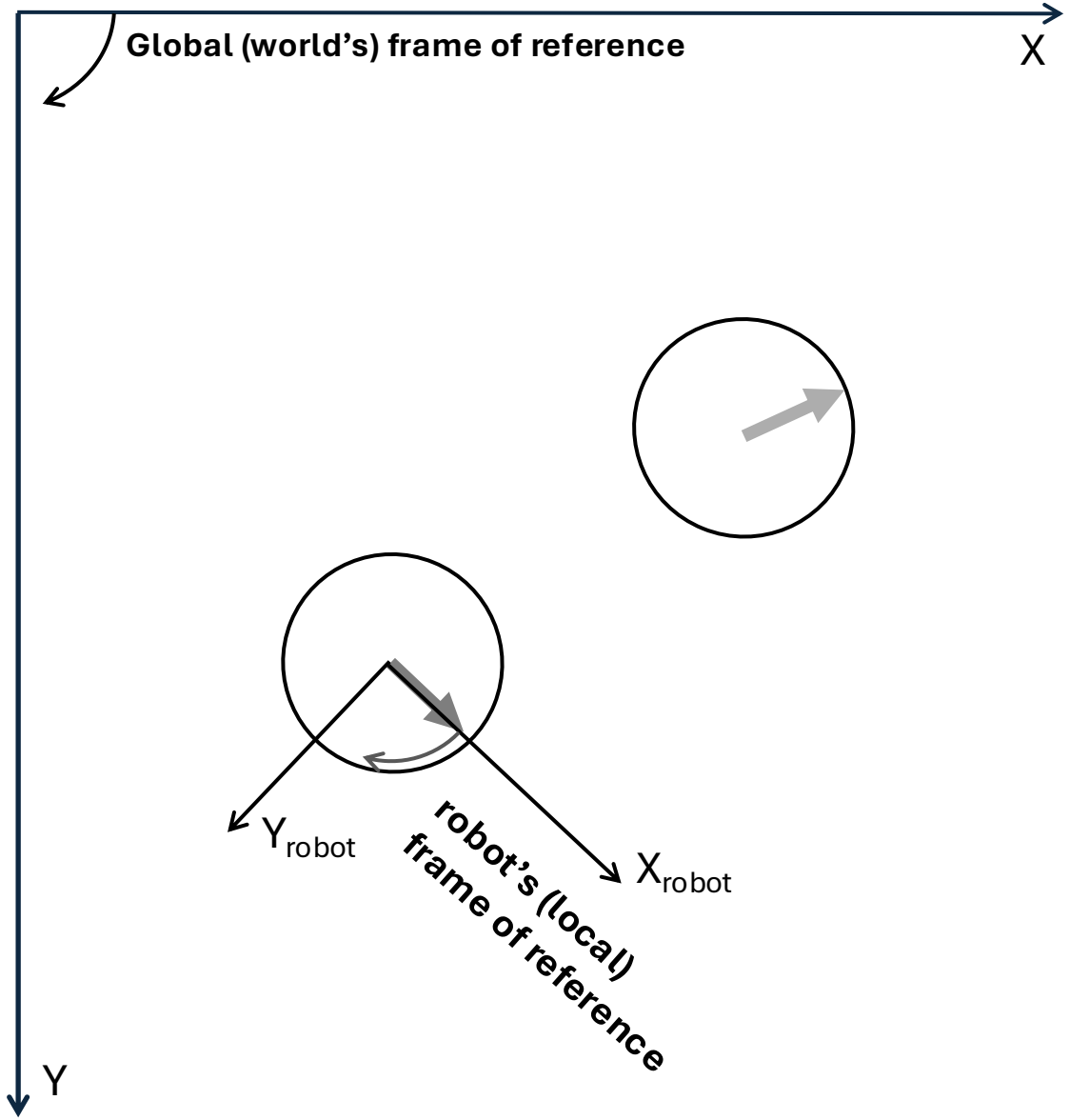
Note:
These robots have an IMU → know their own heading in the world's frame,
But no GPS or other localization systems → don't know their own position in the world's frame

…

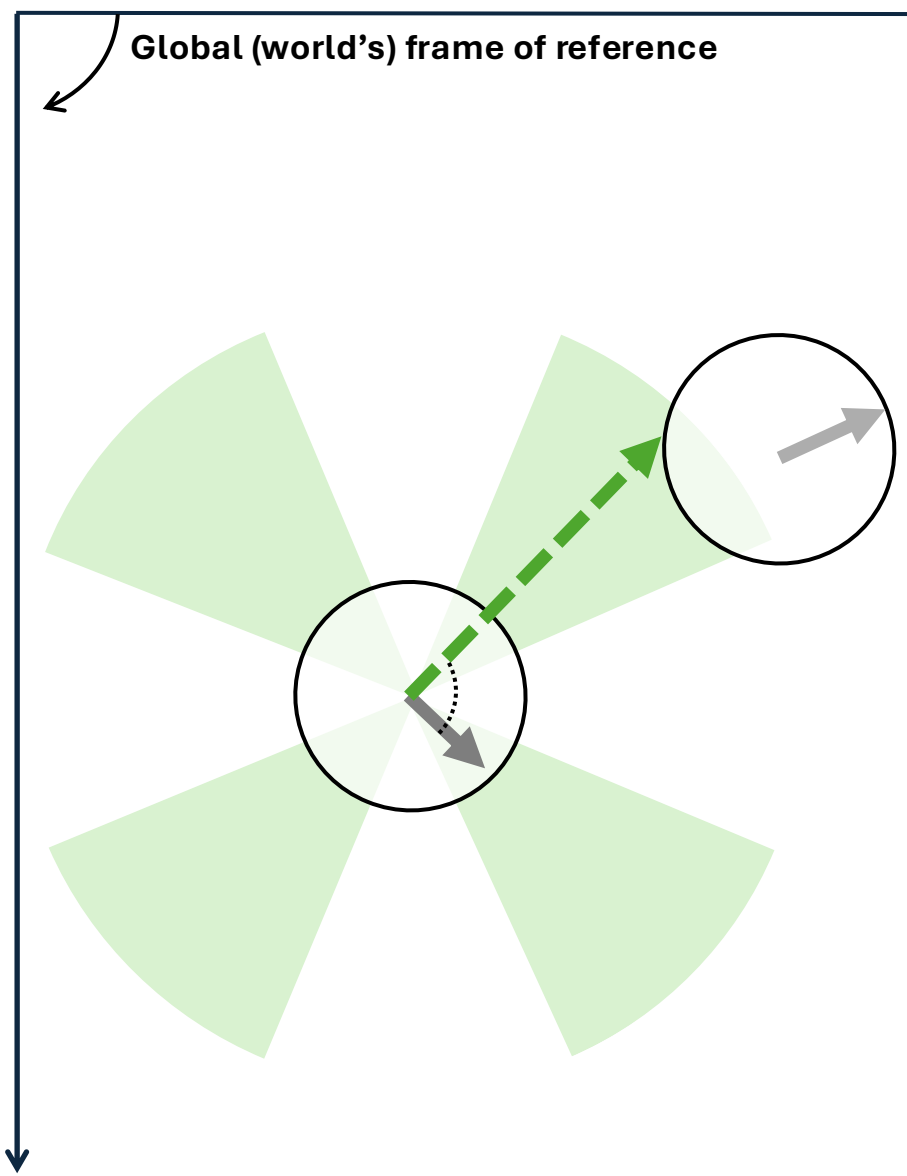
A robot and its peer



A robot and its peer



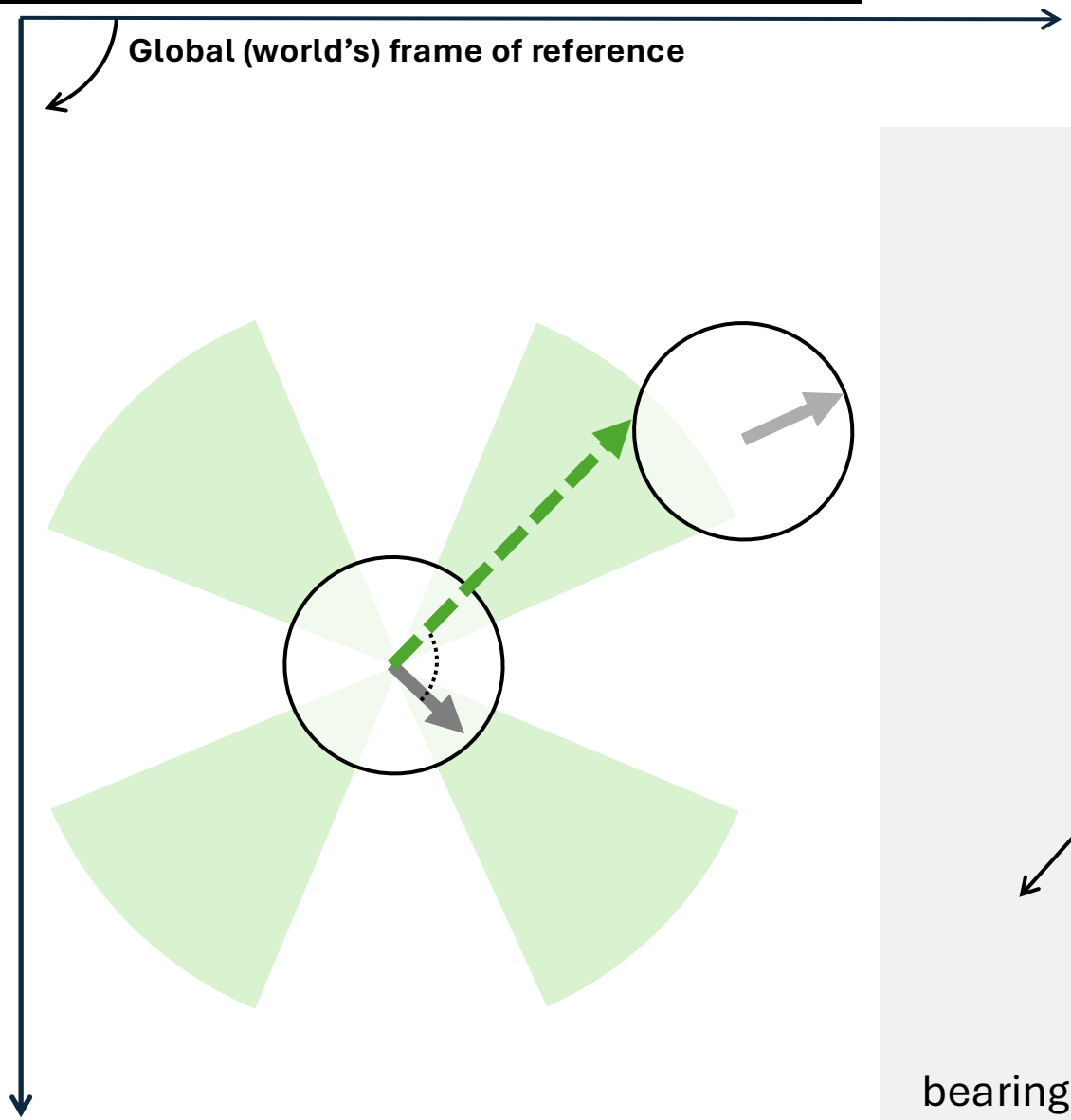
A robot can detect its peer

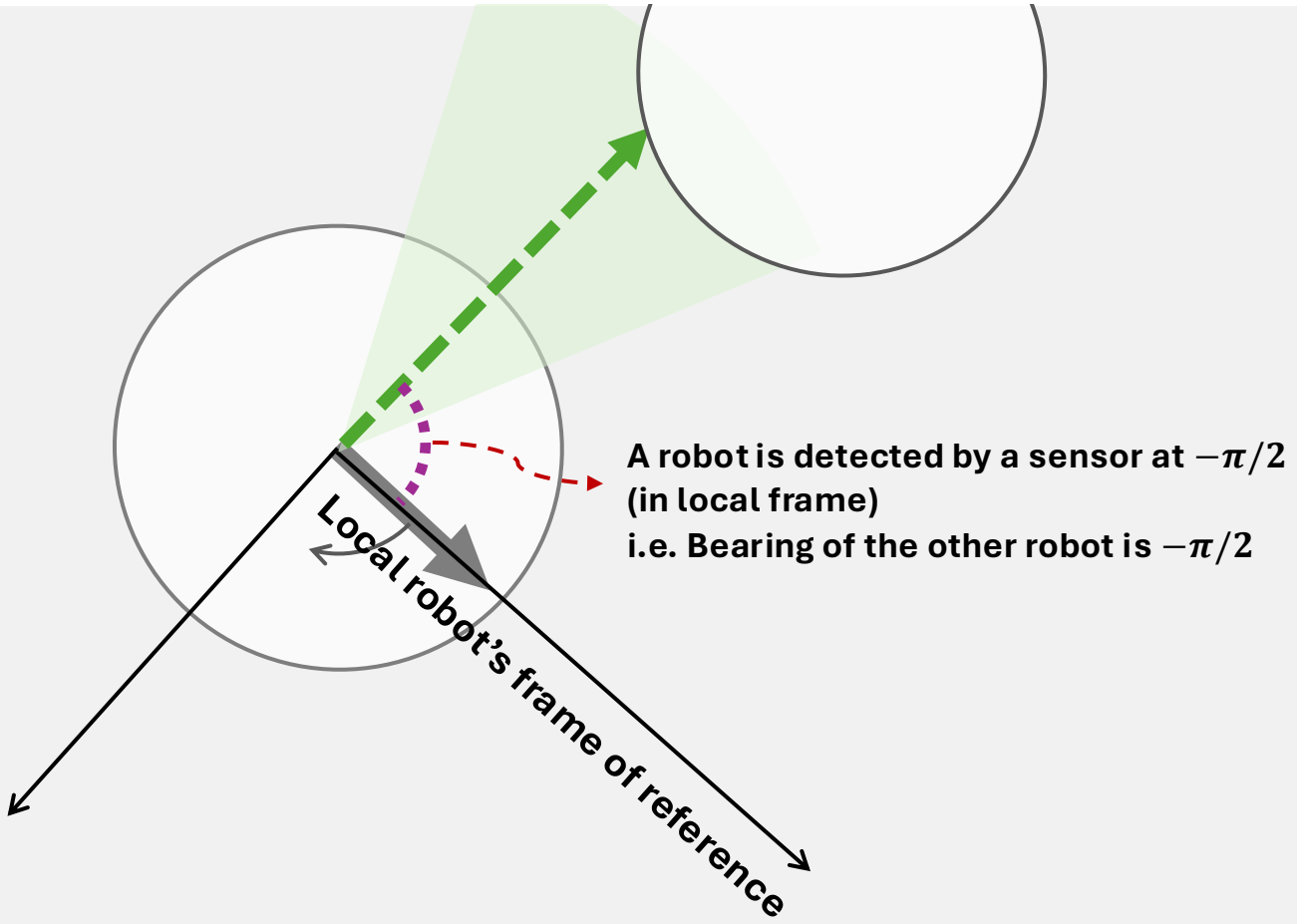


One robot detects another robot within its sensors range.

(in this example we have 8 RAB sensors; different from the simulator)

Other robot's bearing





bearing of a detected object is the angular direction of the object relative to the robot's heading (its frame of reference)

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- Messages received from other robots (e.g., their heading) → **World's frame** of reference

…

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Messages to broadcast:
robot's heading (according to IMU)

Controller algorithm and commands to actuators:
set_rotation_and_speed(angle, speed)